

The weak-coupling gap in a small box reduces to three inequalities through the Schur complement

A lower bound $\Delta \geq \delta_1^{(3)} g^{2/3} (1 - Cg^{2/3}) \hbar c/L$ for the torus Hamiltonian with a momentum cutoff follows from a proved gap-transfer lemma and three inequalities about polynomial operators on finitely many oscillators. The lemma: if P is an orthogonal projection, $D = \bar{P} H \bar{P} \geq \mu$ on $\text{ran } \bar{P}$, and the Schur-complement error $\eta = \sup_{E \leq a_1} \|(D - E)^{-1/2} \bar{P} H P\|^2$ satisfies $a_1 - \eta < \mu$, then the two lowest eigenvalues of H obey $a_0 - \eta \leq \bar{E}_0 \leq a_0$ and $E_1 \geq a_1 - \eta$, so $\text{gap}(H) \geq \text{gap}(PHP) - \eta$. Applied with P the projection onto the ground state of the nonzero momentum modes, the three inequalities are: (H1) exciting any nonzero mode costs at least $(2\pi\hbar c/L)(1 - Cg^{2/3})$ in the presence of the constant modes and the couplings; (H2) the coupling between the constant-mode sector and the excited nonzero-mode sector has Schur error $\eta \leq Cg^{4/3}\hbar c/L$; (H3) the effective constant-mode Hamiltonian PHP , which is C133's operator plus the nonzero-mode zero-point energy as a function of the constant modes, has $\text{gap} \geq \delta_1^{(3)} g^{2/3} (1 - Cg^{2/3}) \hbar c/L$. The scaling analysis of the cubic and quartic couplings shows why each inequality is expected with the stated powers, and identifies the smallness parameter as $g^{2/3}$, Lüscher's expansion parameter. The lemma is proved; the three inequalities are the open content of the lower side of the small-volume corner. Units: $\hbar = c = 1$ in this note, energies in units of $\hbar c/L$ and g dimensionless, as in G07 Proposition 7. Nothing here is promoted.

1. The gap-transfer lemma

Let H be self-adjoint, bounded below, with compact resolvent on a Hilbert space $\mathcal{H}$, and P an orthogonal projection with $P D(H) \subset D(H)$; write $\bar{P} = 1 - P$, $A = PHP|_{\text{ran } P}$, $D = \bar{P} H \bar{P}|_{\text{ran } \bar{P}}$, $B = PH\bar{P}$. Let $a_0 \leq a_1 \leq \dots$ be the eigenvalues of A and $E_0 \leq E_1 \leq \dots$ those of H .

Lemma 1. Suppose $D \geq \mu$ and, for some $\eta \geq 0$, $\|(D - E)^{-1/2} B^*\|^2 \leq \eta$ for all $E \leq a_1$, with $a_1 < \mu$. Then

$$a_0 - \eta \leq E_0 \leq a_0, \quad E_1 \geq a_1 - \eta, \quad E_1 - E_0 \geq (a_1 - a_0) - \eta.$$

Proof. For $E < \mu$, $D - E$ is invertible and the Schur complement of $H - E$ with respect to the decomposition $\text{ran } P \oplus \text{ran } \bar{P}$ is $F(E) = A - E - B(D - E)^{-1} B^*$ on $\text{ran } P$. The block factorization

$$H - E = \begin{pmatrix} 1 & B(D - E)^{-1} \\ 0 & 1 \end{pmatrix} \begin{pmatrix} F(E) & 0 \\ 0 & D - E \end{pmatrix} \begin{pmatrix} 1 & 0 \\ (D - E)^{-1} B^* & 1 \end{pmatrix}$$

is a congruence, so by Sylvester's law of inertia (the number of negative eigenvalues, finite here because H has compact resolvent and $E < \mu \leq \inf D$) the number of eigenvalues of H below E equals the number of negative eigenvalues of $F(E)$, since $D - E > 0$ contributes none. The hypothesis gives $0 \leq B(D - E)^{-1} B^* \leq \eta$, hence $A - E - \eta \leq F(E) \leq A - E$ and, by min-max, the number of negative eigenvalues of $F(E)$ lies between $\#\{k : a_k < E\}$ and $\#\{k : a_k < E + \eta\}$. Taking E slightly above a_0 shows $E_0 \leq a_0$; taking $E = a_0 - \eta$ shows no eigenvalue of H lies below $a_0 - \eta$; taking $E = a_1 - \eta$ shows at most one eigenvalue of H lies below $a_1 - \eta$, so $E_1 \geq a_1 - \eta$. $\square$

The lemma transfers a gap from the compressed operator PHP to H at the price η , which is the size of the coupling between $\text{ran } P$ and its complement measured against the energy $D - E$ needed to leave $\text{ran } P$. Everything below is the choice of P and the estimation of $a_1 - a_0$, μ and η .

2. The cutoff torus Hamiltonian and the projection

On T_L^3 in temporal gauge, with $SU(2)$ for definiteness, momentum cutoff $|k| \leq \Lambda$ and the convention of G07 Proposition 7 ($S = -\frac{1}{4g^2} \int F^a F^a$), decompose $A_i(x) = a_i + \tilde{A}_i(x)$ into the constant

mode and the nonzero modes $\tilde{A}_i(x) = L^{-3} \sum_{0 < |k| \leq \Lambda} \hat{A}_i(k) e^{ik \cdot x}$, with conjugate momenta p_i and $\tilde{\Pi}_i$. Then

$$H = H_0 + H_\perp + W,$$

$$H_0 = \frac{g^2}{2L^3} \sum_i |\vec{p}_i|^2 + \frac{L^3}{2g^2} \sum_{i < j} |\vec{a}_i \times \vec{a}_j|^2, \quad H_\perp = \sum_{0 < |k| \leq \Lambda} \left[\frac{g^2}{2} |\hat{\Pi}(k)|^2 + \frac{1}{2g^2} |k \times \hat{A}(k)|^2 \right]$$

(in the normalization where the free part has frequencies $|k|$), and W collects every term of $\frac{1}{2g^2} \int |F_{ij}|^2$ that is neither purely constant-mode nor quadratic in $\tilde{A}$ at $a = 0$:

$$W = W_2 + W_3 + W_4, \quad W_2 = \frac{1}{g^2} \int (\partial_i \tilde{A}_j - \partial_j \tilde{A}_i) \cdot ([a_i, \tilde{A}_j] - [a_j, \tilde{A}_i]) + \frac{1}{2g^2} \int |[a_i, \tilde{A}_j] - [a_j, \tilde{A}_i]|^2,$$

$$W_3 = \frac{1}{g^2} \int (\partial_i \tilde{A}_j - \partial_j \tilde{A}_i + [a_i, \tilde{A}_j] - [a_j, \tilde{A}_i] + [a_i, a_j]) \cdot [\tilde{A}_i, \tilde{A}_j], \quad W_4 = \frac{1}{2g^2} \int |[\tilde{A}_i, \tilde{A}_j]|^2 - (\text{its zero-mode part also})$$

(The constant-mode part of $\int |[\tilde{A}, \tilde{A}]|^2$ is not separated further; it belongs to W_4 and is quartic in the nonzero modes.) The Gauss constraint is imposed on states; the constant gauge transformations act on a_i by the adjoint action and on the nonzero modes covariantly, and the nonconstant ones are fixed by keeping the transverse nonzero modes, the standard small-volume gauge choice.

Let $\Omega_\perp$ be the ground state of $H_\perp$, the Gaussian vacuum of the free transverse modes, with energy $E_\perp = \sum_k |k|$ (a constant that cancels from gaps) and first excitation energy $\mu_\perp = 2\pi/L$. Set

$$P = 1_{\text{const}} \otimes |\Omega_\perp\rangle\langle\Omega_\perp|.$$

P commutes with the constant gauge transformations because $\Omega_\perp$ is invariant under them, so Lemma 1 can be applied inside the physical sector. The compression is

$$A = PHP = H_0 + \langle\Omega_\perp, W\Omega_\perp\rangle(a) + E_\perp,$$

where $\langle\Omega_\perp, W_2\Omega_\perp\rangle(a)$ is a function of the constant modes only (the Gaussian expectation of a quadratic form in $\tilde{A}$ with a -dependent coefficients), $\langle W_3 \rangle$ vanishes (odd in $\tilde{A}$), and $\langle W_4 \rangle$ is a constant. Thus A is C133's operator plus a potential $U(a) = \langle W_2 \rangle(a)$, which is the leading change of the nonzero-mode zero-point energy with the constant background.

3. The three inequalities and their expected sizes

Rescale $\vec{a}_i = g^{2/3} L^{-1} \vec{\xi}_i$ and $\hat{A}(k) = g \hat{A}'(k)$ so that $H_0 = g^{2/3} L^{-1} h_3(\xi)$ with h_3 the unit operator of C133 and $H_\perp$ is g -independent in the primed variables. Then, counting powers of g at fixed L and at typical values $|\xi| = O(1)$, $|\hat{A}'| = O(1)$:

term	order in units of $1/L$	role
H_0	$g^{2/3}$	the gap to be transferred
$H_\perp$ excitations	$\geq 2\pi$	the energy μ of leaving $\text{ran } P$
W_2 , linear in a	$g^{2/3}$	enters $U(a)$ and B
W_2 , quadratic in a	$g^{4/3}$	enters $U(a)$
W_3 , term with $\partial \tilde{A}$	g	enters B
W_3 , terms with a	$g^{5/3}, g^{7/3}$	enters B
W_4	g^2	enters A and D

With these sizes the three inequalities that Lemma 1 needs are:

(H1) $D = \bar{P} H \bar{P} \geq E_\perp + \frac{2\pi}{L} (1 - C_1 g^{2/3})$ on $\text{ran } \bar{P}$: any state with an excited nonzero mode costs at least one quantum $2\pi/L$ up to the shift produced by W in a background $|a| \sim g^{2/3}/L$.

(H2) $\eta = \sup_{E \leq a_1} \|(D - E)^{-1/2} \bar{P} W P\|^2 \leq C_2 g^{4/3}/L$. The leading contribution is the W_2 term linear in a (order $g^{2/3}/L$, creating one or two nonzero quanta) squared and divided by $2\pi/L$; the W_3 term of order g/L contributes g^2/L .

(H3) $\text{gap}(A) = \text{gap}(H_0 + U(a)) \geq \delta_1^{(3)} g^{2/3} (1 - C_3 g^{2/3})/L$. $U(a)$ is even in a (charge conjugation), its linear term vanishes, and its quadratic term is of order $g^{4/3}/L$, a relatively bounded perturbation of h_3 in the rescaled variables.

Given (H1)–(H3), Lemma 1 with $\mu = E_\perp + \frac{2\pi}{L}(1 - C_1 g^{2/3})$ and $a_1 - \eta < \mu$ (true for small g since $a_1 - a_0 = O(g^{2/3}/L)$ and $a_0 - E_\perp = O(g^{2/3}/L)$) yields

$$\Delta \geq \delta_1^{(3)} g^{2/3} \frac{\hbar c}{L} (1 - C g^{2/3}),$$

restoring $\hbar c$. This is the lower side of the small-volume corner S, matching the upper side obtained in the Polyakov-average note up to pure numbers, and it is the leading term of Lüscher's expansion (abstract, B78) as a two-sided bound.

4. What the reduction shows

The smallness parameter is $g^{2/3}$ in every entry of the table, which is the reason the small-volume expansion is a series in $g^{2/3}$. The mechanism transferred is the zero-point confinement of C133: A is that operator plus a small even potential, and the nonzero modes enter only through the price η of leaving their vacuum. The reduction also shows where the small-volume regime ends: (H1) fails when $C_1 g(L)^{2/3}$ reaches order one, that is when $\Delta_0(L) \sim 2\pi/L$, which is the crossover $z \simeq 2$ of Lüscher–Münster in the variable $z = \Delta L$. Beyond it, leaving the nonzero-mode vacuum is no longer expensive compared with the constant-mode gap, no projection of the form P separates scales, and the argument gives nothing, which is the statement of T2's difficulty in this language.

5. Consequence for STATE

The weak-coupling lower bound is now three inequalities on explicit polynomial operators, with a proved transfer lemma. The next step on the lower side is (H2), the Schur error of W_2 and W_3 against the free nonzero-mode gap, which is a Gaussian computation with the C133 ground state as weight; (H1) and (H3) are relatively bounded perturbation statements. Their proof would close the small-volume corner S from both sides and make the crossover statement the whole remaining content of T2' at fixed cutoff.